

Notes: Edelen (JFE, 1999)

Investor Flows and the Assessed Performance of Open-End Mutual Funds

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1 Big picture

Edelen (1999) revisits a standard conclusion in the mutual fund performance literature: open-end mutual funds appear to earn negative abnormal returns and to exhibit negative market timing. The paper argues that this conclusion is incomplete because it ignores the core service provided by open-end funds: investors can add and withdraw money at low direct cost, and fund managers must trade to accommodate those flows.

The key idea is simple. Flow-induced trades are liquidity-motivated trades. In rational expectations models with informed and uninformed traders, uninformed liquidity trades lose money on average to informed traders. If open-end fund managers are forced to trade because investors buy or redeem fund shares, then measured fund returns combine two objects: the manager's discretionary portfolio-choice ability and the costs of providing liquidity to investors.

The paper's central finding is that this liquidity service is economically important. A unit of liquidity-motivated trading, defined as trading equal to 100% of fund assets per year, is associated with roughly a 1.5–2.0% decline in abnormal returns. Once the paper controls for this cost, the average fund's net abnormal return moves from about -1.6% per year to approximately zero. Negative market timing also disappears once realized flow is included in the timing regression.

2 Incremental contribution of the paper

1. **Reinterprets mutual fund underperformance.** The paper argues that negative average alpha is not necessarily evidence that managers lack skill. It may instead reflect the indirect cost of offering daily liquidity to investors.
2. **Uses fund-level flow and trading data, not only returns.** The N-SAR filings report inflows, outflows, purchases, and sales. This lets the paper measure how much trading activity is plausibly induced by investor flows.
3. **Separates liquidity-motivated trading from discretionary trading.** Total trading is decomposed into a component predicted by investor flows and a residual component interpreted as discretionary trading.
4. **Estimates the performance cost of liquidity provision.** The paper links the flow-induced trading component to subsequent abnormal returns and shows that the coefficient is negative, large, and robust to several return benchmarks.
5. **Addresses reverse causality from returns to flows.** Since flows respond to past performance, a naive regression of alpha on flow is biased. The paper uses lagged abnormal return controls and an instrument based on lagged flow.
6. **Explains negative market-timing estimates.** The apparent negative market timing of mutual funds is shown to occur when, and only when, funds experience flow. Flow interacts mechanically with market movements and induces timing-looking patterns.
7. **Connects fund performance to market microstructure.** The interpretation is grounded in Grossman–Stiglitz type logic: informed trading profits must be offset by losses borne by liquidity traders. Open-end funds partly act as such liquidity traders on behalf of investors.

8. **Changes the appropriate benchmark.** The benchmark for evaluating open-end fund managers should not be a passive portfolio with zero flow cost. It should allow for negative abnormal return generated by realized liquidity demand.

3 Research question and answer

The paper asks:

When evaluating open-end mutual funds, how much of apparent underperformance is actually the cost of satisfying investor flows?

The answer is: a large amount. The average fund in the sample has a conventional annual alpha of about -1.63% . After adjusting for flow-induced liquidity trading, the conditional alpha at zero flow is close to zero. This does not imply that open-end funds create large net value for investors; rather, it implies that managers roughly earn back fees before flow costs, while the liquidity feature absorbs the observed performance deficit.

4 Conceptual mechanism

4.1 Open-end funds provide two services

Open-end mutual fund managers do at least two things at the same time. First, they choose portfolios. Second, they provide liquidity to investors who can add and withdraw capital. Standard performance evaluation often treats the observed return as if it only reflects the first service. Edelen argues that this is a mistake because the second service changes the portfolio and forces trading.

4.2 Why flow-induced trades can lower alpha

Suppose a fund receives redemptions. The manager must raise cash by selling securities. Suppose a fund receives large inflows. The manager must invest cash to avoid drifting away from the fund's target risk exposure. In either case, some trades are made because investors demand liquidity, not because the manager has information.

In a market with asymmetric information, such uninformed liquidity trades are costly. They may pay bid-ask spreads, brokerage commissions, price impact costs, and adverse-selection costs. Therefore, higher flow-induced trading should be associated with lower abnormal returns.

A stylized decomposition is:

$$\alpha_{jt} = \text{manager information component}_{jt} - \text{liquidity-trading cost}_{jt} + \text{noise}_{jt}.$$

The empirical test asks whether the estimated liquidity-trading cost term is negative and large enough to explain the usual negative alpha.

4.3 Why flow can create apparent negative market timing

Negative market timing means the fund's beta appears to fall when the market performs well or rise when the market performs poorly. Edelen's mechanism says this can arise mechanically from flows. If investor flows are correlated with market returns, and if funds adjust exposure only after receiving flows, then measured returns will look as if managers timed the market poorly even if the manager had no negative timing skill.

This is why the market-timing tests interact the timing regressor with realized flow. Once this flow interaction is included, the negative timing coefficient shifts from the manager-timing term to the flow term.

5 Data

The sample contains 166 open-end mutual funds selected randomly from the Summer 1987 edition of Morningstar's *Sourcebook*. Each fund has at least some emphasis on equity securities. On average, the sample contains about five years of data for each fund, with most observations from 1985 to 1990.

Fund flow and trading data are hand-collected from N-SAR filings with the Securities and Exchange Commission. The N-SAR filings report monthly inflows and outflows, and semiannual purchases and sales. Return data are taken mostly from Morningstar, with remaining returns hand-collected from *Barron's*.

The data are especially useful because they include both sides of flow and both sides of trading. This makes it possible to ask whether inflows map into purchases and outflows map into sales, rather than relying only on net flows.

6 Empirical design

6.1 Step 1: estimate the amount of flow-induced trading

The paper first estimates how trading activity responds to flows. The core idea is that if inflows and outflows induce purchases and sales, then flow is a useful proxy for liquidity-motivated trading. The paper estimates regressions such as purchases on inflows, sales on outflows, turnover on cash turnover, and net purchases on net inflows.

The coefficient is around 0.6–0.8. This means that about 60–80 cents of each dollar of flow generates incremental trading. The coefficient is less than one because funds can buffer flows with cash, cross inflows against outflows, or combine flow trades with trades they would have made anyway.

6.2 Step 2: separate liquidity trading and discretionary trading

For fund j at time t , Edelen constructs a proxy for liquidity-motivated trading:

$$\hat{F}_{jt} = \hat{c}(f_{jt}^I + f_{jt}^O),$$

where f^I is inflow, f^O is outflow, and $\hat{c}$ is the estimated flow-trading response coefficient from the trading regressions. A typical value is $\hat{c} \approx 0.70$.

Discretionary trading is then total trading less estimated liquidity-motivated trading:

$$\hat{d}_{jt} = q_{jt}^P + q_{jt}^S - \hat{F}_{jt}.$$

The key return regression is a monthly cross-sectional Fama–MacBeth style regression of abnormal return on estimated liquidity trading, discretionary trading, and lagged abnormal returns.

6.3 Step 3: address reverse causality

Flows respond to returns. Table 4 shows that inflows are strongly related to lagged abnormal returns. Therefore, if a fund has high past alpha and then receives flows, a naive regression could confuse return-chasing behavior with the cost of flow.

The paper addresses this by including 6 or 12 lags of abnormal returns and by instrumenting current estimated liquidity trading with lagged flow. This is designed to isolate the predictable flow component that is not simply contemporaneous return chasing.

6.4 Step 4: test market timing with and without flow interactions

The paper estimates Treynor–Mazuy and Henriksson–Merton style timing regressions. The baseline regression finds negative market timing. The augmented regression includes an interaction between flow and the market-timing regressor. If negative timing is due to flow, the manager timing coefficient should move toward zero or become positive, while the interaction term should be negative. Table 8 shows exactly this pattern.

7 Main findings

7.1 Flow is large enough to matter

The typical open-end fund experiences substantial investor movement. Annual inflows average 65% of assets, annual outflows average 48%, and annual cash turnover averages 33%. Trading is also very large: purchases average 113% of assets, sales average 103%, and turnover averages 90%.

This is the foundation of the paper. If flows were small, liquidity provision could not explain performance. But the data show that flows and trading are both large.

7.2 Flows induce actual trading

A dollar of inflow is associated with about 63–67 cents of purchases; a dollar of outflow is associated with about 68–76 cents of sales. Thus investor flows are not merely accounting entries. They translate into portfolio trades and therefore can affect returns.

7.3 Liquidity trading has negative abnormal return effects

The coefficient on liquidity-motivated trading is consistently negative. In the main specification using net abnormal returns, the coefficient ranges from about -1.74 to -2.23 . Interpreted annually, trading equal to fund assets is associated with roughly a 1.7–2.2 percentage point lower abnormal return.

Adding back expenses and brokerage commissions reduces the magnitude, but the coefficient remains negative. This indicates that out-of-pocket costs matter, but they do not fully explain the effect. Adverse-selection and price-impact costs are likely important.

7.4 The average fund does not underperform after liquidity adjustment

The conventional average annual alpha is -1.63% . After applying the liquidity-adjusted benchmark, the conditional expected alpha at zero flow is about -0.26% using 6 lags and $+0.11\%$ using 12 lags. For the median fund, the adjusted alpha is positive, roughly $+0.20\%$ to $+0.43\%$.

The interpretation is that fund managers' discretionary portfolio choices approximately offset their fees before liquidity costs. The observed negative net performance is largely attributable to the liquidity service.

7.5 Negative market timing is a flow artifact

The standard timing regressions show negative timing. However, once the interaction between flow and the market-timing regressor is included, the negative timing coefficient is absorbed by the flow interaction. In other words, funds look like bad market timers when they experience flow.

8 Tables

8.1 Table 1. Characteristics of the sample of 166 open-end funds

Read. Table 1 describes the sample of 166 open-end funds. The average fund has \$273 million in assets, a median size of \$98 million, total expenses of 1.72% per year, brokerage commissions of 0.44% per year, and a market-model beta of 0.90. The average market-model intercept is -1.63% per year, with a cross-sectional standard deviation of 4.50%.

Interpretation. This table establishes the conventional puzzle. Using the usual market model, the average fund underperforms by about 1.6% per year, close to the fund's reported expense ratio. At first glance, this seems to support the standard view that active managers add little or no value before fees. The paper's contribution is to show that this interpretation is incomplete because the alpha is measured after liquidity-motivated trading costs.

Why it matters. The sample looks reasonably representative of open-end equity funds along size, fees, beta, and return performance. Therefore, the subsequent liquidity-cost adjustment is not applied to a strange sample with unusually extreme funds.

8.2 Table 2. Characteristics of flow and trading activity

Read. Table 2 shows the scale of flows and trading. Inflows average 65% of assets annually; outflows average 48%; net inflow averages 16%. Purchases and sales average 113% and 103% of assets, respectively. Cash turnover is 33%, and trading turnover is 90%. Inflows are more persistent than outflows, with AR(1) coefficients of 0.71 and 0.47.

Interpretation. This is the empirical starting point for the whole paper. Flows are not small relative to assets or trading volume. A large share of trading is mechanically related to investor activity. The turnover cash/trading ratio of 62% on average, and 40% at the median, implies that investor flows account for a large fraction of total trading activity.

Economic message. Open-end fund liquidity is not free. Investors can redeem or add capital cheaply at the individual level, but the portfolio must absorb the aggregate trading cost. Table 2 shows that this cost channel is large enough to plausibly move measured performance.

Table 1
Characteristics of the sample of 166 open-end funds

The data are taken from each fund's N-SAR filing with the Securities and Exchange Commission. The beginning and ending dates of the sample vary across funds, with a mean beginning date of May 1985 and a mean ending date of July 1989. The market model regressions use the CRSP value-weighted index. The indicated variable is first averaged across all observations for a particular fund. Statistics are then presented on these 166 mean values. Total expenses exclude brokerage commissions. Brokerage commissions represent the total payment and the brokerage commission rate represents brokerage commissions divided by the volume of trading.

	Mean	Std. deviation	Median	Time-series std. deviation
First year of operations	1967	16yrs.	1969	
Assets managed (\$millions)	273	433	98	28.2%
Number of monthly obs.	54	12	54	
Total expenses	1.72%	1.60%	1.41%	0.55%
Brokerage commissions	0.44%	0.47	0.32	
Brokerage commission rate	0.21%	0.15	0.17	
Market-model intercept	-1.63%	4.50%	-0.68%	3.00%
Market-model beta	0.90	0.24	0.92	

Table 1. Characteristics of the sample of 166 open-end funds

Table 2
Characteristics of flow and trading activity

All variables are scaled by the average assets managed over the six-month N-SAR filing period. Flow is observed monthly and annualized (multiplied by 12). Trading activity is observed every six months and annualized (multiplied by two). Cash (or trading) turnover is the minimum of the inflow (purchases) over the six-month filing period and the outflow (sales) over that period, annualized (multiplied by two). Net flow is the month's inflow minus outflow, annualized (multiplied by 12). Net purchases are determined at each six-month observation, then annualized (multiplied by two).

Panel A. Sample statistics

The indicated variable is first averaged across all observations for a particular fund. Statistics are then presented on these mean values. Ratios are at the individual fund six-month level, then averaged.

	Mean	Std. deviation	Median	Time-series std. deviation
<i>Flow</i>				
Inflow	65%	72%	38%	67%
Outflow	48	51	34	35
Cash turnover	33	43	22	25
Net inflow	16	48	01	70
<i>Trading activity</i>				
Purchases	113	78	89	54
Sales	103	75	82	42
Turnover	90	65	73	38
Net purchases	10	35	01	46
<i>Ratios</i>				
Inflow/Purchases	61	46	47	
Outflow/Sales	66	53	52	
Turnover: cash/trading	62	57	40	

Panel B. Autocorrelations

Observations are monthly for flow and semiannual for trading activity. The panel presents the coefficient estimate from an AR(1) model.

	Flow	Trading
Inflow	0.71	Purchases 0.21
Outflow	0.47	Sales 0.15
Cash turnover	0.67	Turnover 0.17
Net inflow	0.49	Net purchases 0.13

Panel C. Marketwide components

The panel presents the average regression statistics across funds. Per-fund inflow (outflow) is regressed on concurrent aggregate inflow (outflow) (source of aggregate data: Investment Company Institute).

	Coefficient	R ²
Inflow	0.54	0.12
Outflow	0.53	0.13

Table 2. Characteristics of flow and trading activity

8.3 Table 3. Regressions of trading activity on flow

Read. Table 3 estimates whether flows lead to trades. In the time-series regressions, purchases respond to inflows with a coefficient of 0.63, and sales respond to outflows with a coefficient of 0.76. In the cross-sectional regressions, the corresponding coefficients are 0.67 and 0.68. Turnover and net-purchase regressions produce similar evidence.

Interpretation. For each dollar of investor inflow or outflow, roughly 60–80 cents generates incremental security trading. The coefficient is below one because managers use cash buffers, offset inflows against redemptions, and sometimes combine flow trades with discretionary trades. But the coefficient is far from zero, so flow is a powerful proxy for liquidity-motivated trading.

Magnitude. The time-series estimates imply liquidity-motivated purchases of 24% of assets and liquidity-motivated sales of 28% of assets per year. Scaled by total trading volume, these components are roughly 29% and 27%. This means the liquidity service accounts for a very large part of what fund managers do operationally.

8.4 Table 4. Flow regressed on 12 lags of abnormal returns

Read. Table 4 regresses inflows and outflows on 12 lags of abnormal returns. Inflows are strongly positively related to lagged abnormal returns across many lags. Outflows are much less related, with only the $t - 12$ coefficient reaching a notable t -statistic of 2.3.

Interpretation. This table documents reverse causality. Investors direct inflows toward funds with good prior abnormal returns. Therefore, a simple negative relation between current flow and current abnormal return is not mechanically guaranteed. In fact, return-chasing would bias the estimated flow-performance effect upward, making it harder to find a negative liquidity-cost effect.

Why it matters for identification. Because flows follow returns, Edelen cannot simply regress alpha on flow. The paper controls for lagged abnormal returns and instruments current flow-induced trading with lagged flow. Table 4 is the justification for that design.

Table 3
Regressions of trading activity on flow

Four groups of regressions are presented, corresponding to (1) purchases τ^P on inflow f^I (2) sales τ^S on outflow f^O (3) trading turnover τ^T on cash turnover f^T (4) and net purchases τ^N on net inflow f^N . The observation length is six months. All variables are scaled by the average fund size over the six-month N-SAR filing period and annualized. The estimated volume of liquidity-motivated trading at the fund is calculated as the regression coefficient times the average of the corresponding flow measure at that fund, scaled by either the fund's average size or average trading volume.

Panel A presents the mean of the indicated statistic averaged over a separate regression for each of 128 funds for which at least eight time-series observations are available (the mean regression degrees of freedom is 7.9). Standard error estimates are the sample standard errors (across funds) of the relevant statistic. Panel B presents the mean of the indicated statistic averaged over a separate regression for each of 12 semi-annual periods (the mean regression degrees of freedom is 122). Standard error estimates are the sample standard errors (across dates) of the relevant statistic. Standard errors are in parentheses.

	Intercept	Coefficient	R^2	Adjusted R^2	Estimated liquidity-motivated trading volume, scaled by	
					Fund size (%)	Trad. volume (%)
<i>Panel A. Time-series regressions</i>						
τ^P on f^I	0.84 (0.08)	0.63 (0.15)	0.31 (3.0)	0.22 (3.0)	24 (0.04)	29 (0.04)
τ^S on f^O	0.71 (0.06)	0.76 (0.17)	0.22 (2.0)	0.11 (2.0)	28 (0.06)	27 (0.05)
τ^T on f^T	0.73 (0.06)	0.74 (0.22)	0.18 (2.0)	0.07 (2.0)	15 (0.04)	18 (0.04)
τ^N on f^N	0.02 (0.10)	0.75 (0.06)	0.48 (3.0)	0.42 (3.0)		
<i>Panel B. Cross-sectional regressions</i>						
τ^P on f^I	0.81 (0.04)	0.67 (0.04)	0.30 (3.0)	0.29 (4.0)	36 (0.04)	33 (0.04)
τ^S on f^O	0.74 (0.07)	0.68 (0.08)	0.16 (2.0)	0.14 (3.0)	30 (0.04)	30 (0.03)
τ^T on f^T	0.77 (0.06)	0.55 (0.07)	0.13 (2.0)	0.10 (3.0)	18 (0.03)	21 (0.03)
τ^N on f^N	0.75 (0.10)	0.75 (0.10)	0.55 (3.0)	0.55 (4.0)		

Table 3. Regressions of trading activity on flow

Table 4
Flow regressed on 12 lags of abnormal returns

A cross-sectional regression is run each month for which sufficient data are available (66 months). Two sets of regressions are presented, with a dependent variable equal to the individual fund monthly cash inflow and outflow scaled by the average assets managed over the corresponding six-month N-SAR filing period. The regressors are the fund's abnormal return for month $t-1$ through $t-12$ from a time series regression of fund returns on the CRSP value-weighted index. All variables are annualized. The table presents the average of the coefficient estimates across the 66 regressions (t -statistics in parentheses).

	AR:	$t-1$	$t-2$	$t-3$	$t-4$	$t-5$	$t-6$	$t-7$	$t-8$	$t-9$	$t-10$	$t-11$	$t-12$
Inflow	0.25 (3.0)	0.25 (2.8)	0.25 (3.0)	0.16 (2.2)	0.25 (3.4)	0.15 (2.0)	0.17 (2.5)		0.12 (1.6)	0.15 (2.1)	0.04 (0.5)	0.15 (2.3)	0.21 (3.0)
Outflow	0.01 (0.2)	0.08 (1.1)	0.08 (1.1)	0.01 (0.1)	-0.01 (-0.1)	0.01 (-0.1)	0.00 (0.0)		-0.01 (0.1)	0.00 (0.0)	0.01 (0.1)	0.02 (0.4)	0.14 (2.3)

Table 4. Flow regressed on 12 lags of abnormal returns

8.5 Table 5. Abnormal return regressed on flow and discretionary trading

Read. Table 5 is the main alpha result. The coefficient on liquidity-motivated trading is negative across specifications. Using net returns and a univariate market model, the coefficient is -1.74 with six lags and -2.23 with 12 lags. The estimates remain negative when expenses or brokerage commissions are added back, and the quadratic market model produces similar results.

Interpretation. A flow-induced trading volume equal to 100% of assets is associated with roughly a 1.7–2.2 percentage point decline in annual abnormal returns. This is the paper's central empirical result. The observed negative performance of open-end mutual funds is strongly related to the volume of liquidity-motivated trading they must conduct.

Expenses versus adverse selection. Adding back fund expenses and brokerage costs reduces the coefficient, but does not remove it. This suggests that the liquidity cost is not just explicit fees or commissions. The remaining negative coefficient is consistent with price impact and adverse selection: liquidity trades are executed against better-informed market participants.

Discretionary trading. The coefficient on discretionary trading is generally insignificant. This does not necessarily imply managers have no information. The discretionary-trading proxy is noisy and may contain other noninformational trades such as rebalancing and tax trades.

8.6 Table 6. Abnormal return regressed on separate components of flow and discretionary trading

Read. Table 6 repeats the alpha analysis using inflow, outflow, and net flow separately. The inflow/purchase coefficients are strongly negative and statistically meaningful. Outflow/sale coefficients are negative but noisier. Net-flow coefficients are also negative, especially with 12 lags.

Interpretation. The liquidity-cost effect is not only a redemption story. It appears when funds receive inflows and have to buy securities, and it also appears in the expected direction when funds face outflows and have to sell securities. The strongest precision comes from inflows because inflows are more variable and persistent in the data.

Connection to Table 5. Combining inflows and outflows in Table 5 produces a cleaner signal because both buy-side and sell-side liquidity trades should be costly. Table 6 verifies that the sign of the effect is not driven by one arbitrary aggregation choice.

Table 5
Abnormal return regressed on flow and discretionary trading

The following cross-sectional regression is estimated for each of the 66 months in the sample:

$$AR_{jt} = a + \lambda(\hat{F}_{jt}) + \delta\hat{d}_{jt} + \sum_{i=1}^K \eta_{i,t} AR_{jt-i} + e_{jt}$$

where

- $\hat{F}_{jt}$ is the proxy for liquidity-motivated trading equal to $\hat{f}_{jt-1} \sum_{i=1}^{66} \hat{b}_i$, where $\hat{b}_i$ is the estimated coefficient in a cross-sectional regression of flow ($\hat{f}_{jt} = \hat{c}(f_{jt}^b + f_{jt}^s)$) on lagged flow, $\hat{c}$ is the average flow-trading response coefficient from Table 3, and $f_{jt}^b(f_{jt}^s)$ is the gross inflow (outflow).
- $\hat{d}_{jt}$ is the estimated discretionary trading, equal to the total trading (fund j , period t) less $\hat{F}_{jt}$.
- AR_{jt} is the abnormal return of the fund. Four measures of abnormal returns are considered, corresponding to *net* (fund returns net of all expenses except sales charges); *net + expenses* (net plus adding back the reported total expenses charged to the fund); *net + brok*Volume* (net plus the brokerage commission rate times the trading volume of the fund); and *net + expenses*Volume*.
- The abnormal return is calculated from a regression of the indicated return measure on the CRSP value-weighted index (Panel A) or on the index excess return and its square (Panel B).

The table presents the average (t -statistic) of the coefficient estimates across the 66 cross-sectional regressions. The intercept and coefficient estimates on the lagged-abnormal return control variables are not presented. (The intercept is always insignificant. The average coefficient is 0.02 and they are individually significant out to nine lags.) Results are presented with $K = 6$ and $K = 12$ (t -statistics in parentheses).

	Six lags of abnormal returns		Twelve lags of abnormal returns	
	λ	δ	λ	δ
<i>Panel A. Abnormal returns from a univariate market model</i>				
<i>Net</i>	-1.74 (-2.4)	-0.18 (-0.9)	-2.23 (-3.1)	-0.16 (-0.8)
<i>+ expenses</i>	-1.37 (-1.9)	0.11 (0.5)	-1.77 (-2.4)	0.07 (0.4)
<i>+ brok*Volume</i>	-1.43 (-2.2)	0.03 (0.1)	-1.92 (-2.7)	0.10 (0.5)
<i>+ expenses</i>	-1.11 (-1.5)	0.26 (1.2)	-1.27 (-1.8)	0.30 (1.4)
<i>Panel B. Abnormal returns from a quadratic market model</i>				
<i>Net</i>	-1.90 (-2.7)	-0.16 (-0.8)	-2.11 (-2.9)	-0.13 (-0.7)
<i>+ expenses</i>	-1.37 (-2.1)	0.11 (0.6)	-1.40 (-2.1)	0.08 (0.4)
<i>+ brok*Volume</i>	-1.55 (-2.1)	0.04 (0.2)	-1.54 (-2.4)	0.07 (0.3)
<i>+ expenses</i>	-1.01 (-1.3)	0.27 (1.4)	-0.95 (-1.5)	0.27 (1.4)

Table 5. Abnormal return regressed on flow and discretionary trading

Table 6
Abnormal return regressed on separate components of flow and discretionary trading

This table is similar to Table 5, Panel A (abnormal returns from a univariate market model) except that the regressions use a different measure of flow and discretionary trading. Gross inflow is used in Panel A ($\hat{f}_{jt} = \hat{c}f_{jt}^b$), gross outflow is used in Panel B ($\hat{f}_{jt} = f_{jt}^s$), and net inflow is used in Panel C. Discretionary trading is purchases, sales, and net trading, respectively, less the corresponding flow measure. (t -statistics in parentheses).

	Six lags of abnormal returns		Twelve lags of abnormal returns	
	λ	δ	λ	δ
<i>Panel A. Inflow</i>				
<i>Net</i>	-3.10 (-2.5)	-0.18 (-0.5)	-3.67 (-3.1)	-0.11 (-0.3)
<i>+ expenses</i>	-2.22 (-1.9)	0.44 (1.0)	-2.67 (-2.5)	0.43 (1.1)
<i>+ brok*Volume</i>	-2.77 (-2.2)	0.26 (0.6)	-2.84 (-2.8)	0.40 (1.0)
<i>+ expenses</i>	-1.55 (-1.3)	0.73 (1.8)	-1.73 (-1.66)	0.83 (2.0)
<i>Panel B. Outflow</i>				
<i>Net</i>	-2.15 (-0.9)	-0.76 (-1.7)	-3.60 (-1.8)	-0.64 (-1.5)
<i>+ expenses</i>	-1.00 (-1.0)	-0.20 (-0.5)	-2.99 (-1.5)	-0.25 (-0.6)
<i>+ brok*Volume</i>	-2.29 (-1.0)	-0.42 (-0.8)	-3.17 (-1.6)	-0.3 (-0.7)
<i>+ expenses</i>	-1.69 (-0.7)	0.03 (0.1)	-2.21 (-1.1)	0.11 (0.2)
<i>Panel C. Net flow</i>				
<i>Net</i>	-2.29 (-1.2)	-0.15 (-0.8)	-3.57 (-1.9)	-0.10 (-0.5)
<i>+ expenses</i>	-1.80 (-0.9)	0.17 (0.9)	-3.25 (-1.6)	0.13 (0.7)
<i>+ brok*Volume</i>	-2.96 (-1.4)	0.11 (0.5)	-4.22 (-2.1)	0.12 (0.6)
<i>+ expenses</i>	-1.17 (-0.6)	0.32 (1.6)	-2.52 (-1.3)	0.32 (1.6)

Table 6. Abnormal return regressed on separate components of flow and discretionary trading

8.7 Table 7. Abnormal return regressed on flow and discretionary trading with a multifactor return model

Read. Table 7 repeats the main alpha test using a multifactor benchmark with the CRSP value-weighted index plus size, dividend-yield, and lagged-return factors. The liquidity-trading coefficient

remains negative. For net returns, it is -1.48 with six lags and -1.93 with 12 lags.

Interpretation. The main result is not an artifact of using a simple one-factor market model. Controlling for size, dividend yield, and momentum-related benchmark returns attenuates the coefficient modestly but does not eliminate it. The liquidity-cost channel survives a richer benchmark.

Why this is important. Mutual fund performance estimates are sensitive to benchmarks. If the liquidity result disappeared under a multifactor benchmark, the paper’s interpretation would be fragile. Table 7 shows that the result is robust to a more flexible model of expected returns.

8.8 Table 8. Flow and market-timing measures of performance

Read. Table 8 estimates market-timing regressions. Without flow interactions, the squared-return timing coefficient is -0.24 with a t -statistic of -2.9 , and the absolute-value timing coefficient is -0.033 with a t -statistic of -4.2 . After adding flow interactions, the direct timing coefficient becomes positive or close to zero, while the flow interaction is negative and statistically meaningful, especially for lagged flow.

Interpretation. Standard timing tests make funds look like negative market timers. But the negative timing pattern is actually concentrated in periods when funds experience flows. This means the bad timing result is generated by the liquidity service, not by systematic manager forecasting errors.

Connection to the paper’s thesis. Table 8 is the market-timing analogue of Table 5. The same logic that explains negative alpha also explains negative timing: investors’ liquidity demands force funds to adjust exposure at bad times relative to the benchmark test.

Table 7
Abnormal return regressed on flow and discretionary trading with a multifactor return model

Table 7 differs from Panel A of Table 5 only in that abnormal returns are calculated using a multifactor benchmark. In addition to the CRSP value-weighted index, the factors are:

- The return differential on a portfolio of the smallest size-decile stocks over the largest size-decile stocks
- The return differential on a portfolio of the highest dividend-yield stocks
- The return differential on a portfolio of the previous best-performing stocks over the previous worst performing stocks

(t -statistics in parentheses).

	Six lags of abnormal returns		12 lags of abnormal returns	
	λ	δ	λ	δ
Net	-1.48 (-2.3)	-0.31 (-1.8)	-1.93 (-2.8)	-0.26 (-0.8)
+ expenses	-1.05 (-1.5)	-0.11 (-0.5)	-1.41 (-2.0)	-0.03 (-0.1)
+ <i>brok</i> *Volume	-1.26 (-1.8)	-0.17 (-0.9)	-1.73 (-2.3)	0.07 (0.2)
+ expenses	-0.90 (-1.2)	0.06 (0.4)	-1.00 (-1.6)	0.11 (0.5)

Table 7. Abnormal return regressed on flow and discretionary trading with a multifactor return model

Table 8
Flow and market-timing measures of performance

Regressions are run separately for each fund using the 135 funds with at least thirty monthly observations. The table presents averages of the statistics from these individual-fund regressions (the t -statistics in parentheses are calculated using the sample standard error of the 135 individual-fund estimates). The regressions are

$$\tilde{R}_p = \alpha + \gamma_1 \tilde{R}_{Mt} + \gamma_2 \tilde{X}_{Mt} + e_{pt}$$

$$\tilde{R}_p = \alpha + \gamma_1 \tilde{R}_{Mt} + \gamma_2 \tilde{X}_{Mt} + \gamma_3 (\hat{c}(f^1 + f^0)) \tilde{X}_{Mt} + e_{pt}$$

$\tilde{R}_p$ ($\tilde{R}_{Mt}$) denotes the fund (CRSP value-weighted) return. The market-timing regressor $\tilde{X}_{Mt}$ is either $\tilde{R}_{Mt}^2$ or $|\tilde{R}_{Mt}| \cdot \hat{c}$ is the estimated flow-trading response coefficient taken from Table 3 and f^i (f^0) is the gross inflow (outflow). Flow is either concurrent with returns or lagged one month. All returns are at an annual rate (t -statistics in parentheses).

Market-timing regressor	Squared excess returns				Absolute value excess returns			
	α	γ_1	γ_2	γ_3	α	γ_1	γ_2	γ_3
	-0.08 (-0.2)	0.89 (44)	-0.24 (-2.9)		-0.56 (1.0)	0.90 (44)	-0.033 (-4.2)	
concurrent flow	-0.21 (-0.4)	0.89 (44)	0.23 (1.1)	-0.59 (-2.4)	0.21 (0.4)	0.90 (44)	0.041 (0.2)	-0.037 (-2.0)
Lagged flow	-0.31 (0.3)	0.89 (44)	0.45 (2.1)	-0.82 (-3.4)	0.15 (0.3)	0.90 (44)	0.018 (1.1)	-0.057 (-3.4)

Table 8. Flow and market-timing measures of performance

9 How to read the empirical logic

The paper’s argument has a clear chain:

1. Open-end funds face large investor inflows and outflows.

2. These flows cause real trading in the underlying portfolio.
3. Liquidity-motivated trading is costly in asymmetric-information markets.
4. Therefore, measured fund returns should decline with flow-induced trading.
5. Once that cost is removed, average abnormal performance is no longer negative.
6. Once flow is included in market-timing regressions, negative market timing disappears.

The empirical contribution is that every link in this chain is measured. Table 2 measures the size of flows. Table 3 measures the mapping from flows to trades. Table 4 handles the return-to-flow reverse-causality problem. Tables 5–7 measure the return cost. Table 8 shows the same flow mechanism in market-timing tests.

10 Important limitations

- **The sample is relatively small.** The main sample has 166 funds, with most data from the mid-1980s to 1990. The evidence is strong but historically specific.
- **N-SAR data are hand-collected and semiannual for trading.** Flow is monthly but trading is semiannual. This creates measurement noise, especially for discretionary trading.
- **The liquidity-trading proxy is estimated, not directly observed.** The paper infers liquidity-motivated trading from the flow-trading relation. This is reasonable but not equivalent to observing trade motives.
- **Discretionary trading is a noisy residual.** It includes many trades that may not be informational, so the insignificant discretionary coefficient should not be overinterpreted as proof that managers have no skill.
- **The instrument relies on lagged flow.** The identifying assumption is that lagged flow predicts current liquidity trading but is not correlated with the nonpersistent component of current abnormal returns after controlling for lagged abnormal returns.
- **The paper evaluates funds net of sales loads.** It focuses on open-end fund portfolio returns net of expenses but not on every investor-level cost of entering and exiting funds.

11 Key takeaways

Edelen (1999) changes the interpretation of mutual fund underperformance. Open-end funds do not only select securities; they also absorb investors’ liquidity shocks. That liquidity service imposes real trading costs, and those costs show up as negative alpha and negative market timing in conventional performance tests.

The core takeaway is not that active management is highly profitable. It is that the proper benchmark for an open-end fund is not a passive portfolio with no flow obligations. The benchmark should reflect the indirect cost of providing liquidity. Once this adjustment is made, the average fund no longer appears to underperform on a net-of-expenses basis.

For research design, the paper is also a good example of using institutional detail to reinterpret an asset-pricing anomaly. The data on inflows, outflows, purchases, and sales allow the author

to distinguish a genuine manager-performance signal from a mechanical cost of the mutual fund organizational form.